

Question description

Rathik organized technical round interview in Macrosoft for the set of computer science candidates.

The problem is to perform Implement a stack using single queue. you have to use queue data structure, the task is to implement stack using only given queue data structure.

Rathik have given the deadline of only 5 minutes to complete the problem.

Can you Help the candidates to complete the problem within the specified time limit ?

Function Description

x is the element to be pushed and s is stack

push(s, x)

- 1) Let size of q be s.
- 2) Enqueue x to q
- 3) One by one Dequeue s items from queue and enqueue them.

Removes an item from stack

pop(s)

1) Dequeue an item from q

Constraints

$0 < n, m < N$

$1 < \text{arr}[i] < 1000$

Input Format:

First line indicates n & m, where n is the number of elements to be pushed into stack and m is the number of pop operation need to be performed

next line indicates the n number stack elements

Output Format:

First line indicates top of the element of the stack

second line indicates the top of the element after the pop operation

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 2
1 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 3
```

Test Case 2

INPUT (STDIN)

```
15 4
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
```

EXPECTED OUTPUT

```
top of element 5
top of element 11
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void Stack::push(int val)

Test Case 4

KEYWORD

q.pop();

Test Case 2

KEYWORD

q.push(val)

Test Case 3

KEYWORD

void Stack::pop()

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

60

Code Editor

c

1#include <stdio.h>

2int main()

3{

4

Custom Input (stdin)

T1

T2

Type Here

Stack9.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 1000
5
6 int q[MAX];
7 int front = 0;
8 int rear = 0;
9
10 void q_push(int val) {
11     q[rear++] = val;
12 }
13
14 int q_pop() {
15     return q[front++];
16 }
17
18 int q_size() {
19     return rear - front;
20 }
21
22 int q_front() {
23     return q[front];
24 }
25
```

```
26 void push(int val) {  
27     int s = q_size();  
28     q_push(val);  
29     for (int i = 0; i < s; i++) {  
30         q_push(q_pop());  
31     }  
32 }  
33
```

main | CH.SC.U4CSE25225 | Stack9.c

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```
35     if (q_size() > 0) {  
36         q_pop();  
37     }  
38 }  
39
```

```
40 int top() {
41     if (q_size() > 0) {
42         return q_front();
43     }
44     return -1;
45 }
46
47 int main() {
48     int n, m;
49     if (scanf("%d %d", &n, &m) != 2) return 0;
50
51     for (int i = 0; i < n; i++) {
52         int val;
53         scanf("%d", &val);
```

```
54     push(val);
55 }
56
57 printf("top of element %d\n", top());
58
59 for (int i = 0; i < m; i++) {
60     pop();
61 }
62
63 printf("top of element %d\n", top());
64 return 0;
65}
66
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
5 2
```

```
1 2 3 4 5
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack9.exe
```

```
top of element 5
```

```
top of element 3
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
15 4
```

```
9 8 7 6 5 5 6 7 8 3 11 2 3 4 5
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Stack9.exe
```

```
top of element 5
```

```
top of element 11
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```